Blind Audit of J-Lens / R-Lens / Logit-Lens on Qwen3.6-27B

Executive Summary (DRAFT - agent synthesis, provisional). Final exec summary & pivot are the human's.

1. Question and hypotheses

Does a lens's signal in a blind scan survive when the text-reconstruction channel is neutralized? Three competing hypotheses are discriminated:

- H1 - internal readout: the lens surfaces a genuine internal signal, beyond surface text.
- H2 - text inversion: the apparent signal is just the scan letting a reader reconstruct the text.
- H3 - judge apophenia: the judge manufactures 'anomalies' from anomaly-shaped noise.

2. Method

11 human-validated anomalous/clean text pairs (families: bug, false_premise, injection). Each pair is scanned by 3 instruments (J-lens, R-lens, logit lens) into per-position, per-layer top-k token stacks. Two LLM judges (judgeA = claude-sonnet-4.6; judgeB = gpt-4.1-mini) score each scan for anomaly + confidence, blind to any anomaly family or instrument name. Seven fixed conditions include the controls that carry the argument: prompt-only (text baseline), blind scan, scan_permuted (positions shuffled = H3 control), and reconstruction (judge A rebuilds text from the scan, judge B reads only that = H2 control). A leak audit gates every scan run.

3. Headline results (all figures/numbers computed this session; see Result.md)

Leak gate GREEN for 4-GRAMS only (0 shared 4-grams across 66 scans). CAVEAT: the scan echoes the anomaly token itself at its own position ~100% of the time (see 4b); 9 of 11 items have 1-3 token anomalies, so a MONO-token leak is not excluded and is the relevant channel here. 579/592 valid verdicts (2.2% loss). Independent from-scratch AUC recompute matches the pipeline to <1e-6.

- Text baseline is near-ceiling: prompt-only AUC 0.95 (judgeA) / 0.96 (judgeB).
- Pooled blind-scan AUC per instrument: logit 0.84 >= rlens 0.80 >= jlens 0.75 (n=11/11).
- Triplet (dAUC, 95% paired-bootstrap CI), pooled over judges/prompts:
- D1 scan - prompt: jlens -0.21 [-0.47,0.01] | rlens -0.16 [-0.30,-0.02] | logit -0.12 [-0.30,0.05]
- D2 scan - reconstruction: jlens +0.11 [-0.30,0.47] | rlens +0.49 [0.23,0.79] | logit +0.32 [0.05,0.57]
- D3 permuted - chance: jlens +0.28 [0.02,0.46] | rlens +0.20 [0.01,0.39] | logit +0.29 [0.11,0.45]
- Inter-judge agreement (scan, v1): Cohen kappa 0.49, Spearman 0.50 (moderate).
- Evidence fidelity: ~52% of tokens the judge CITES are absent from the scan; ~13% of the present ones fall in the anomaly zone. High false-alarm on CLEAN twins (reconstruction 0.5-1.0).

4. What the data asserts (factual)

- D3 > 0 with CI excluding 0 on all three instruments: shuffling positions does NOT destroy the discriminative signal -> a large part of it is order-invariant.
- D1 < 0 everywhere: the blind scan never beats reading the text.
- The logit lens - which has no privileged internal-readout machinery - gives the HIGHEST blind-scan AUC, matching/beating the two workspace lenses.
- The judge fabricates the majority of its cited evidence and flags clean twins often.

4b. Content control - a one-token perturbation propagates (key null NOT run)

Does the anomaly leave an objective fingerprint in the scan (not just via the judge)? On the 5 length-aligned single-token pairs, compare the anomalous vs clean scan top-k per position, split by region relative to the anomaly at position s. Under causal attention only positions >= s can be affected; positions > s carry the SAME input token in both twins, so any difference there is the anomaly's internal contextual propagation. Fraction of (position, layer) cells that differ:

- upstream (<s): 0.000 / 0.000 / 0.000 (early/mid/late) - causal sanity check PASSED (exactly 0).
- at (=s): 0.956 / 1.000 / 1.000 - trivial: the input token itself differs (echo). Ignore.
- downstream (>s): 0.254 / 0.602 / 0.720 - SAME input token -> internal propagation of the anomaly.

So a one-token change alters the scan at downstream positions (identical input there) in 25% early / 60% mid cells. IMPORTANT (post-review): this has NO matched null. Under causal attention ANY one-token substitution (even a correct/neutral one) produces this - so it is a fact about transformers, not a measure of anomalies. The proper control (re-scan with a neutral substitution) needs the GPU and was NOT run; it is DECLARED as the key open control. The rise with depth fits residual-perturbation diffusion (not concept computation, which peaked at ~L24 in validation), and 'differ' partly reflects rank churn in flat late-layer distributions.

Anomaly-specificity (free test, 5 pairs): downstream of the anomaly, WHICH tokens does the anomalous scan add? For the semantic-error families the added vocabulary shifts toward the anomaly: 'Incorrect / wrong / error / correct' (+ the Chinese word for 'wrong') appear downstream of fp_02, fp_03 and bug_05 - across ALL three lenses (logit included). Suggestive that the model represents 'something is wrong', not arbitrary reshuffle. Caveats: noisy (~12 tokens/cell), n=5, still no null, still NOT lens-specific.

5. Bugs and anomalies encountered (and fixed)

- judgeB (gpt-5-mini) reasoning-truncated EVERY JSON verdict (finish_reason=length, empty) -> switched to gpt-4.1-mini (non-reasoning, still OpenAI family).
- Judge client had no timeout -> a stalled call hung the whole run; max_tokens=600 too low for claude on verbose scans -> added timeout=90 and raised to 1200.
- content_filter: claude refuses to read some reconstructed texts, biased onto false_premise (~10 lost keys) -> makes D2 fragile.
- Windows cp1252 vs UTF-8 crash on every scan read -> fixed at the root (encoding='utf-8').

6. Limitations / caveats

- n = 11 pairs: pooled CIs are wide; per-family AUCs are exploratory noise.
- The permutation control preserves token CONTENT (only order is shuffled), so D3>0 cannot separate apophenia (H3) from a real-but-position-invariant readout. This is the key design gap.
- D2 compares the scan to a BROKEN reconstructor (high false-alarm, degenerate outputs, content_filter hole) -> it cannot cleanly reject H2.
- Judge B was changed mid-project (gpt-5-mini -> gpt-4.1-mini); results differ from the frozen plan.

7. Provisional conclusion (AGENT OPINION - to be verified, not the pivot)

Honest, deflationary reading (after review). The judge-side result is the robust part: blind-scan AUC is modest, never beats reading the text, is HIGHEST for the 'dumb' logit lens (per-family on v1: logit bug 0.91 > jlens 0.61 / rlens 0.66), shows order-invariant behaviour (D3), fabricates most cited evidence, and false-alarms on clean twins - so whatever is detected is not specific to the J/R-lens machinery and the judge reads it noisily/apophenically. On the instrument side, the content control only shows that a one-token change propagates downstream - which is trivially true for ANY substitution and lacks the neutral-substitution null (not run, needs GPU). The one non-trivial hint is that for semantic-error families the downstream vocabulary shifts toward 'incorrect/wrong/error' (across all lenses) - suggestive that the model represents the error, but noisy, n=5, null-less, and lens-agnostic. Net: no evidence of a lens-SPECIFIC internal readout; a plausible but unproven generic 'error is represented' signal; the audit's value is in the controls and in naming its own open null.

8. Potential leads / next steps

- Content control DONE and POSITIVE (see 4b): the anomaly propagates internally. Next, test whether the downstream change is anomaly-INTERPRETABLE (not just any reshuffle) - e.g. does an anomaly-trained probe read it, or does the surfaced vocabulary shift toward the anomaly semantics.
- Foreign-scan control (judge a different clean item's scan) to bound the JUDGE-side apophenia directly, now that the scan is known to contain real signal.

- Fix or drop the reconstruction channel before trusting D2 (content_filter + degeneracy).
- Classify ALL false positives (not just the 5 in the seed-0 sample) into (a) evokes family / (b) evokes own content / (c) invents; proportion (c) is the direct H3 estimate.
- Scale n beyond 11 if budget allows; report only pooled AUC + triplet as primary.

9. Budget and artifacts

Spend 14.97/19. Artifacts:
results/{metrics,triplet,inter_judge,leak_check,evidence_check,family_false_alarm,blockH_worksheet}, figs/fig1_triplet.png, fig2_auc_family.png, judge/outputs/verdicts.jsonl, Result.md (full record), experiments.md entry (14).

Figure 1 - H1/H2/H3 triplet per instrument (95% paired-bootstrap CI)

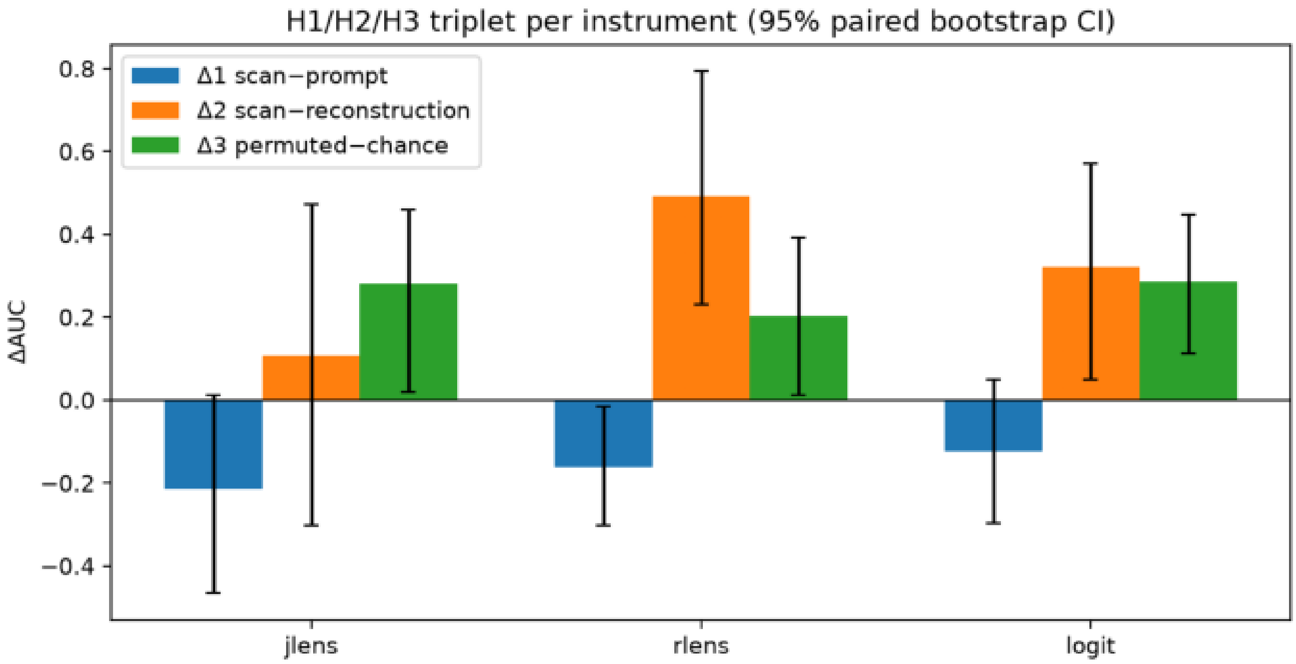

Figure 2 (CORRECTED, v1 only) - AUC per family x instrument. The submitted v1+v2 version inverted the logit ranking via n=2 v2 cells; on v1 logit's best family is bug (0.91), not false_premise.